# From Alignment to Reasoning: A Survey of Technology in Detecting Hateful Memes

Q.Q. Chen (2330034001), S.P. Chen (2330026016), G.C. Jiang (2330034021)
M.H. Li (2330034027), Y.R. Liang (2330034030), Y.X. Tu (2330034048)
Beijing Normal-Hong Kong Baptist University

***Abstract*—Hateful memes represent a significant challenge for online content moderation, as their harmful intent often emerges from the complex interaction between images and overlaid text, rather than from either modality in isolation. This multimodal nature makes if difficult for traditional single-modality models and basic fusion techniques to handle the problems, as they struggle to decode nuanced meanings like sarcasm or contextual hate content This literature review comprehensively surveys the technologies for hateful meme detection, beginning with data preprocessing techniques such as OCR-based text extraction and image inpainting for images with text. It then explores advanced model encoders like Perception Encoder, which are designed to align visual and textual semantics. The core of the survey focuses on methods for improving model interpretability and realism, such as attention matrices and text and modules that explicitly model false claims or semantic contradictions. These techniques enables fine-grained cross-modal reasoning to identify implicit hateful content. By synthesizing these advances, this literature review outlines a pathway toward developing more robust, accurate, and context-aware moderation systems that can better mimic human-like understanding and address the challenges of harmful multimodal content on the internet.**

***Index Terms*—Hateful Meme Detection, Multimodal Learning, Content Moderation, Cross-modal Fusion, Social Media Analysis**

## I. INTRODUCTION

As a new cross-disciplinary field in Natural Language Processing (NLP), the growth of the Internet and the popularity of social media have made memes a common and important form of communication. Modern internet memes usually contain an image with text, expressing various ideas, often in a funny or critical manner. However, their ability to spread quickly and widely also makes them a powerful tool for sharing harmful messages. These memes combine images and text to spread biased views—like racism, sexism, or divisive beliefs—in indirect or hidden ways.

This situation poses unique challenges for NLP. The sheer volume of online content makes manual moderation impractical. Furthermore, the multimodal nature of harmful memes places the task beyond traditional text-based NLP, as the offensive meaning arises from the complex combination of image and text rather than from either modality alone. This creates major hurdles for automated moderation systems.

Early attempts at automated meme moderation, which relied on single-modality models or basic techniques like feature fusion, have proven inadequate. The early moderation systems were too simplistic to decode the nuanced, layered meanings that emerge from the combination of images and text (like sarcasm or memes that only make sense with specific context or knowledge of current events). As a result, they consistently missed this covert hate content. Moreover, existing methods depend heavily on pre-trained models and lack the ability to incorporate real-world context, such as evolving online slang or current eventswhich is a capability that essential for human-like understanding.

Inspired by initiatives like the Hateful Memes Challenge by Facebook, this NLP research project aims to provide a comprehensive review of recent advances designed to address these limitations.

This paper examines the complete technical pipeline for hateful meme detection. The process begins with NLP-based data preprocessing, which includes text extraction via OCR and keyword identification. It then proceeds to the core methodology, covering advanced model encoders (e.g., Perception Encoder) and fine-grained fusion mechanisms like feature interaction matrices. Finally, we explore explainability components, such as attention matrices.

Through this survey of multimodal NLP techniques, this report seeks to outline directions for developing more robust, accurate, and context-aware systems to improve moderation of harmful multimodal content, demonstrating the broad potential of NLP technologies in addressing complex real-world problems.

## II. LITERATURE REVIEW

### A. Background and Definition

A meme is defined as an image accompanied by text that can be widely disseminated through the Internet (1). It generally refers to a combination of picture and text that conveys personal beliefs, often carrying humorous or satirical meaning (2).

With the continuous development of internet and photo-editing technologies, memes have become a popular form of communication on social media. While most memes are harmless, some may contain hateful images or text. On a personal level, it has been shown that exposure to negative memes can negatively affect mental health (3). On a societal level, hateful memes involving racism or secessionism may deepen social divisions. Research also suggests a connection between such memes and real-world criminal behavior (4).

The vast scale of the internet has made manual detection impossible. The multimodal nature of memes further complicates automated detection. To counter the spread of harmful

memes, Facebook launched the "Hateful Memes Challenge" to advance research in this field. Our survey is inspired by this challenge.




Fig. 1: Examples of images in Hateful Meme Challenge

However, first, single models or standard multimodal fusion methods are insufficient. Early fusion, which merges raw inputs, assumes a descriptive relationship between text and image. This assumption is often wrong in real-world memes, since the two modalities may imply different things. Late fusion, which combines the outputs of modality-specific models, may fail to capture the complex interactions between fine-grained, cross-modal attributes (5). Second, existing methods always rely on the pre-trained models, especially pretrained LLMs. They leverage the distinct reasoning apabilities of different models and are optimized for the hateful meme classification task based on their respective design principles. Most importantly , we observe that current methods fail to integrate real-time news and evolving internet slang to identify more implicit hate expressions, nor do they incorporate contextual presuppositions for more human-like comprehension. Therefore, we aim to explore new methods that are better suited for the task of meme moderation.

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### *B. Data Preprocessing Technology*

In multimodal hateful content detection, the raw input typically consists of images, text, and complex visual semantics that jointly shape the overall meaning. To ensure that the downstream model can accurately understand the potential offensive intent conveyed by the interaction between images and text, it is necessary to perform systematic and structured preprocessing before the data enters the main inference framework.

Specifically, the preprocessing pipeline in this study includes text detection and recognition, mask generation for text regions, image inpainting, prompt construction, multi-LLM text fusion, and keyword extraction of critical semantic units. Through these operations, the original meme images are transformed into cleaner and more interpretable multimodal inputs, providing a solid foundation for fairness, robustness, and accuracy in hateful meme detection.

*1) OCR-based Text Extraction*

In the task of hateful meme detection, the substantial portion of offensive meaning is not conveyed by the image alone, but rather emerges from the combination of visual content and overlaid text. This multimodal interplay becomes particularly salient in memes involving sensitive topics such as race, gender, nationality, or religion, where the textual component often carries the core hateful intent.

Given these observations, the first step in assessing whether a meme contains harmful content is to extract all visible text within the image. To do so, we reviewed available text extraction methods and found that Optical Character Recognition (OCR) provides an effective and widely used solution. OCR can automatically convert overlaid meme text into machine-readable form without manual transcription (6).

The OCR module maps an input image $I$ to a set of recognised text units:

$$\text{OCR}(I) = \{(b_i, t_i)\}_{i=1}^{n}, \tag{1}$$

where $t_i$ denotes the extracted textual content. The recognised text contents $\{t_i\}$ can then be directly fed into downstream NLP-based hate speech classification models.

*2) Text-Region Masking and Image Inpainting*

Beyond text extraction, prior multimodal studies highlight the value of obtaining a text-free version of the image to better isolate visual semantics (7). Building on this insight, our review focuses on two commonly paired preprocessing techniques: EAST for detecting text regions and LaMa-based inpainting for restoring natural visual content after text removal.

The Efficient and Accurate Scene Text Detector (EAST) (8) is widely adopted in visual-text understanding pipelines due to its ability to localize text regions directly from convolutional feature maps without the need for complex proposal mechanisms. EAST predicts rotated bounding boxes or quadrilateral shapes through a fully convolutional network, enabling robust detection of irregularly positioned meme text.

$$B = \text{EAST}(I) = \{b_1, b_2, \ldots, b_n\}, \tag{2}$$

where each $b_i$ denotes a predicted text bounding region. To further support downstream reconstruction, these detected regions are converted into a binary mask $M$ indicating the pixels that require inpainting:

$$M(x,y) = \begin{cases} 1, & (x,y) \in \bigcup_i b_i, \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

The Big LaMa framework (9) applies large-mask inpainting using Fourier Convolutions (FFC) and global receptive fields. This design enables the model to reconstruct missing regions with coherent structure, semantic consistency, and smooth texture transitions even under heavy occlusion. Big LaMa preserves global context while recovering fine-grained details, making it well suited for meme preprocessing where text regions are large and irregular.

Given an image $I$ and its corresponding mask $M$, the reconstructed version is obtained through $I_{\text{clean}}$:

$$I_{\text{clean}} = \text{LaMa}(I, M). \tag{4}$$

Our observations show that combining EAST with Big LaMa consistently produces coherent, text-free images. These cleaner images enable more reliable visual feature extraction

for multimodal hate detection. Removing overlaid text before visual encoding also helps separate linguistic cues from visual ones and reduces interference between modalities. As a result, the downstream model receives a clearer multimodal representation. These findings highlight the importance of including text-region detection and inpainting in preprocessing pipelines for robust hateful meme analysis.

*3) Prompt-enhanced Network*

After obtaining text-free images from the inpainting stage, we examined existing captioning and prompt-enhancement methods to generate more task-relevant semantic descriptions. The Prompt-enhanced Network (Pen) (10) improves hateful meme classification by learning optimized prompts and refining feature representations through contrastive training. Pen shows that enhanced prompts can strengthen the ability of pretrained language models, such as RoBERTa, to distinguish subtle differences between hateful and non-hateful expressions. Its multi-view mechanism also suggests that prompt enhancement may help image-derived text descriptions become more sensitive to hateful cues, providing useful guidance for our later model design.

*4) LLM-BLENDER*

Building on this direction, further literature review led us to the cross-model integration framework *LLM-BLENDER* (11). This framework ensembles multiple large language models (e.g., GPT, Gemini, Qwen) using a combination of pairwise ranking and weighted fusion. Instead of relying on simple majority voting or score averaging, LLM-BLENDER employs a learnable aggregation strategy. Given $N$ candidate outputs $\{y_1, \ldots, y_N\}$, the pairwise ranking module first constructs a comparison matrix $M$, where each entry $M_{ij}$ represents the confidence that candidate $y_i$ is better than $y_j$. A weighted score for each candidate is then computed using the MaxLogits function:

$$s_i = \sum (M_i^* - M_i^*), \tag{5}$$

where $M_i^*$ denotes the $i$-th row and $M_i^*$ the $i$-th column of the comparison matrix. The top-$K$ ranked candidates are subsequently fused by a generative model to produce a final, more reliable output.

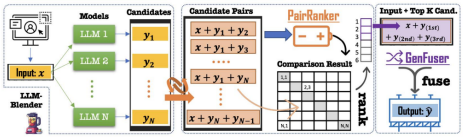



Fig. 2: The LLM-Blender Pipeline

The design of LLM-BLENDER demonstrates the potential of cross-LLM ensemble learning for improving reasoning consistency and reducing single-model bias.

*5) BERT Keyword Extraction*

Through our review of multimodal hateful meme research, we found that key offensive lexical cues play a crucial role in determining whether a meme expresses hateful intent. These cues may include slurs, derogatory terms, group identifiers (e.g., race, gender, religion), or implicit negative expressions, all of which strongly affect how the meme is interpreted. To identify such cues in both (1) OCR-extracted text and (2) LLM-generated descriptions from purified images, recent work commonly uses BERT-based keyword extraction methods such as KeyBERT (12). These methods leverage contextual embeddings to produce more accurate and semantically informed keyword representations.

The typical BERT keyword extraction pipeline (e.g., KeyBERT) first computes contextual embeddings for each token of a sentence $S = \{w_1, w_2, \ldots, w_n\}$:

$$e_i = \mathrm{BERT}(w_i). \tag{6}$$

The overall sentence embedding is then computed as the average of token embeddings:

$$e_S = \frac{1}{n} \sum_{i=1}^{n} e_i. \tag{7}$$

The importance score of each token is obtained via cosine similarity between the token embedding and the sentence embedding:

$$\mathrm{score}(w_i) = \cos(e_i,\, e_S). \tag{8}$$

Tokens with higher scores are considered more semantically salient. Compared with traditional statistical approaches such as TF–IDF (13), BERT-based methods capture richer contextual dependencies and implicit meanings, making them particularly suitable for detecting metaphorical, ironic, or indirect hateful expressions . Based on our review, such methods offer a more fine-grained and discriminative representation of textual signals, which is crucial for downstream hateful meme classification.

Furthermore, recent studies show that large language models can also be used for keyword extraction by guiding them with task-specific prompts (14). These LLM-based approaches benefit from stronger reasoning capabilities and can identify sensitive or context-dependent expressions that traditional keyword extractors may miss. For hateful memes, this is particularly useful because certain harmful cues only emerge when textual phrases are interpreted together with visual context. This insight suggests that we directly call a fine-tuned BERT model In our pipeline and provide it with a task-specific prompt, such as: *"Within the sentence {text}, extract keywords that may cause ambiguity or have a negative effect when combined with the image in the task of hateful meme detection."* This allows the model to identify potentially harmful lexical units in a controlled and reproducible manner.

## C. Core Technology

*1) Model Encoder*

Hateful meme detection is widely regarded as a particularly challenging problem for automatic content moderation, mainly because the offensive meaning is typically constructed through the combination of image and text rather than being contained in either modality alone. Such combinations create implied

negative meanings that require a deep understanding of the relationship between the two modalities. This necessitates an encoder capable of effectively modeling the intricate relationships between visual and textual components to identify potentially harmful content.

*a) CLIP-based Encoder*

To enable effective meme content moderation, accurate recognition of meme content is essential. Modern memes often convey meaning through intricate interactions between visual and textual elements, which presents challenges for conventional simplistic models. We have therefore explored and implemented an approach using CLIP as the core encoder, which serves as the base model of our project.

CLIP is pre-trained on image-text pairs using a dual-encoder architecture. This model effectively aligns visual and textual representations in a shared semantic space, making it well-suited for multimodal content moderation(15).

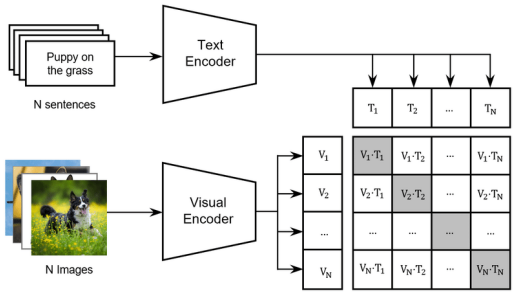



Fig. 3: The structure of CLIP Encoder

The CLIP-based model acquires a shared image-text embedding space by applying contrastive learning to a large amount of image-text pairs. It consists of two main encoders: an image encoder based on the Vision Transformer (ViT) and a text encoder based on a standard Transformer (16). The image encoder processes images to extract visual features, while the text encoder processes text to extract semantic features. Both encoders map their inputs into a shared multimodal embedding space.

The image encoder uses the ViT architecture, which treats an image as a sequence of patches. Specifically, the input images are split into fixed-size and non-overlapping patches. Each patch is linearly projected into an embedding vector. To retain spatial information, learnable position embeddings are added to this sequence.

In addition, a special classification token is prepended to the sequence. This token aggregates global image features in the final output. As shown in Equation (9), the full input sequence is formed by summing up the patch embeddings, the classification token and the position embeddings.

The sequence is then processed by multiple Transformer layers. Each layer contains multi-head self-attention(MSA) and a feedforward network(FNN), with layer normalization and residual connections which goal is to stabilize training. The output corresponding to the classification token from the last layer is used as the global image feature and projected into the multimodal embedding space.

$$z_0 = [x_{\text{class}}, x_{p_1}^E, x_{p_2}^E, \ldots, x_{p_N}^E] + E_{\text{pos}} \tag{9}$$

The text encoder uses a standard Transformer. The input text is first tokenized and converted into word embeddings. Then, Position embeddings are added and the sequence is processed by Transformer layers. The output corresponding to the end-of-sequence token is extracted as the global text representation.

We are inspired by the strengths of CLIP encoder in image-text alignment. Its dual-encoder architecture can effectively integrates visual and textual semantics. This enables accurate identification of implicit harmful content, making it suitable for meme moderation.

After reviewing other papers, we also discovered that for a model trained solely through image-text contrastive learning (using the CLIP objective), the features from certain intermediate layers can serve as universal features for a variety of downstream tasks and they are better than that in output layer. Therefore, we consider using the features from the last hidden layer of the encoder before pooling.

*b) Perception Encoder (PE)*

The primary innovation of Perception Encoder (PE) is in its departure from the standard CLIP practice of using only the final output layer. Instead, it deeply explores and integrates features from the intermediate layers of the model. The core is the transformation of intermediate layers from transitional states into a set of powerful and flexible representations that can use in visual tasks (17).

In a standard CLIP encoder, the features generated by intermediate layers are merely a by-product of the training process so it is not a good one for us to use. However, it is known that shallow features primarily capture detailed information such as edges, textures and colors. These features have high spatial accuracy but weak semantics. Mid-level features begin to form local structures, such as eyes or noses. Deep-level features, being closer to the final layer, are rich in global semantic information.

PEcore draws inspiration from the Feature Pyramid Network (FPN) in computer vision, systematically utilizing features from all layers. Consequently, the model produces multi-scale representations (17). For the same image regions, the model can understand what it is at a macro level while also discerning its specific boundaries and textures at a micro level.

Additionally, PE employs cross-attention method at the intermediate layers, which allows visual features to be shaped in real time under the guidance of textual information. As a result, the intermediate visual features can get the language-aware capabilities.

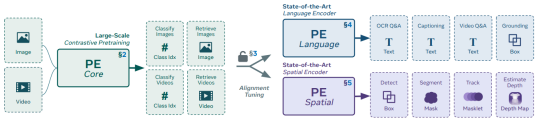



Fig. 4: The function of Perception Encoder (PE)

In terms of training methodology, PEcore significantly improves model performance and robustness under the same computational budget and amount of training data.

The training does not use high resolution from start to finish. Instead, PEcore divides training into multiple stages (e.g., 98px, 154px), starting from low resolution and gradually increasing it. This greatly enhances training efficiency, allowing for longer training within the same computational limit.

Furthermore, it uses a larger batch size with the LAMB optimizer, increasing the batch size from 32K to 64K. This provides the model with more opportunities to encounter hard negative samples within each batch. Also, the LAMB optimizer enables the use of a higher learning rate, which accommodates the changing resolution strategy.

Additionally, PEcore incorporates Rotary Position Embedding in 2D (2D RoPE) into the attention layers (17). This enhances the model's understanding of positional information and its ability to generalize to different resolutions, leading to notable improvements in robustness metrics.

*2) ROI Image Patch Extraction*

Region of Interest (ROI) refers to a specific area within an image, a set of data, or a scene that is specially marked as the focus for analysis, processing, or attention. Simply, ROI is a discrete area within a dataset or scene that is designated for focused analysis or processing. A critical step in determining whether a meme contains hateful content is extracting local evidence from the visual content that is highly semantically relevant to the text (18).

We investigated multiple ROI extraction techniques and ultimately selected a semantic-guided extraction approach based on a multimodal model. We developed this idea through the whole literature review. We aim to leverage the image–text shared embedding space established during CLIP pretraining to enable semantically guided ROI extraction, which is not achievable with traditional purely visual ROI extraction methods based on Faster R-CNN (19). This method overcomes the limitations of traditional purely visual object detection, ensuring that the extracted image patches focus on regions semantically related to potential hateful keywords.

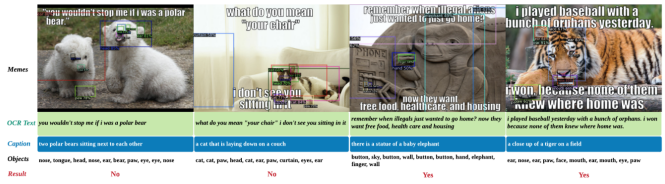



Fig. 5: Qualitative examples of ROI

The purpose of the ROI image patch extraction module is to establish fine-grained associations between abstract concepts in the text and specific pixel regions in the image. It enables precise localization by leveraging semantic consistency to accurately identify and crop the most relevant image patches, avoiding the introduction of redundant background information. Additionally, it constructs an interaction matrix to facilitate the interaction between the extracted features and the features of the textual keywords.

*3) Cross-Modal Fusion*

*a) Feature Interaction Matrixs*

Aligning the image features and text features one by one is a crucial task in the identification of hate expressions. After extracting the ROI features, the model will obtain many resized image features. Match these features with the textual features derived from the analysis of emojis by large language models one by one. This requires cross-model fusion.

In spite of earlier multimodal approaches such as Riza Velioglu et al. conducted a study in 2020 provide valuable insights into combining visual and textual streams, their fusion mechanisms remain relatively coarse (20). The model relies on VisualBERT-style attention to integrate region-level visual features with text, but it does not explicitly model fine-grained cross-modal interactions. The model must depend on implicit attention patterns to infer cross-modal relationships, which makes it less sensitive to subtle semantic contradictions between image and text, and also reduces interpretability.

In the detection of hateful memes, an image can only become hateful when combined with a certain piece of text. Just looking at the image or just reading the text alone cannot determine if something is hateful. Only certain specific combinations of image features and text features can give rise to the meaning of hatred. CLIP conducts contrastive pre-training through approximately 400 million image-text pairs, simultaneously training the image encoder and the text encoder, so that the correct image-text pairs have the highest similarity in the joint embedding space. The downstream tasks are directly performed with natural-language prompts for zero-shot transfer. This paradigm significantly reduces the reliance on labeled data and task-specific heads (15) .




Fig. 6: An example hateful meme with a conflict between the image and the text

However, the images and texts in the pre-training dataset usually have a high degree of similarity, but this is rarely the case in hateful memes. This is because in hateful memes, the content implied by the images and texts may be different.(5) Therefore, cross-modal fusion is required.

After conducting extensive recent research, we have come to understand the **Feature Interaction Matrices** (**FIM**) technology. FIM forms the outer product of the projected image

and text embeddings to model every dimension-wise cross-modal correlation. This explicit, bilinear fusion preserves the fine-grained correspondences that signal hateful intent and also affords clearer interpretability in downstream workflow analysis.

After obtaining two unimodal features, noted as $(f_i, f_t)$, through the CLIP encoder, learnable projection layers are superimposed on them to map the two representations into a task-aligned space that is more suitable for the task. In order to align the two modalities of images and text into the same comparable subspace before the middle-level fusion, and to facilitate subsequent explicit cross-modal interactions, specifically, the study applies two projection layers to $f_i$ and $f_t$ respectively, and explicitly sets the same dimension $n$. That is as shown by the following equation. This way, it not only retains the generalization advantage of CLIP, but also achieves task-oriented alignment with extremely small trainable parameters.

$$p_i = Proj_i(f_i) \in \mathbb{R}^n, p_t = Proj_t(f_t) \in \mathbb{R}^n, \tag{10}$$

From the above process, we obtained $(p_i, p_t)$. As shown in Equation (11), given the image embeddings $p_i \in \mathbb{R}^{n*n}$ and text embeddings $p_i \in \mathbb{R}^{n*n}$ obtained through the projection layers, FIM is defined as their outer product and bilinear pooling.

$$R = p_i \bigotimes p_t \in \mathbb{R}^{n \times n} \tag{11}$$

This enables the explicit encoding of the correlation between each image dimension and text dimension in a straightforward and understandable manner, namely dimension-wise correlations. Unlike the implicit interaction methods of concatenation or simple attetion method like that of Kiela et al. (concatBert)(21), FIM explicitly writes the corresponding combinations of image and text dimensions as triggered combined semantic information into a matrix. This naturally makes the image-text combinations or conflicts more learnable, to facilitate fine-grained interpretability and the detection of unique image-text conflicts and combined semantic contents in hateful memes.

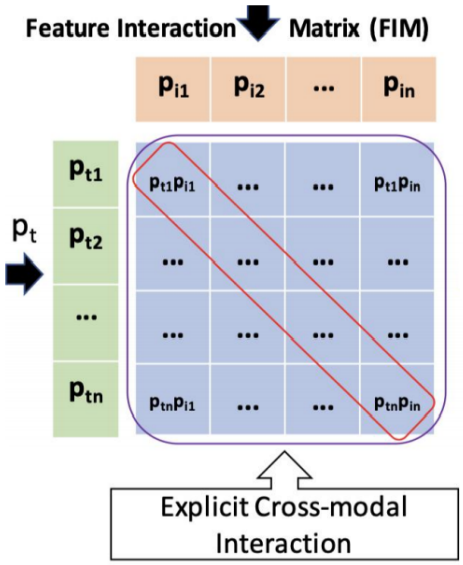



Fig. 7: The Feature Interaction Matrices (FIM)

Furthormore, In order to convert the explicit cross-modal interaction matrix into learnable interaction feature vectors, it need to **flatten** the obtained matrix. Operationally, it construct the FIM as the bilinear outer product

$$R = p_i \otimes p_t \in \mathbb{R}^{n \times n}, \tag{12}$$

and then flatten it as

$$r = \mathrm{vec}(R) \in \mathbb{R}^{n^2}. \tag{13}$$

The flattened interaction vector is passed to downstream classification.

*b) Dynamic Weighted Gating*

To determine whether the combination of pictures and text pairs has an emotional tendency, the first step usually involves identifying and analyzing the textual cues, while also taking into account the visual information. However, in the detection of hateful memes, problems such as modal mismatch and high computational cost often occur: The different modal sampling rates and noise structures lead to the fusion of features without proper alignment, which seriously affects the discrimination; while the pairwise attention mechanism with full alignment results in a significant increase in computational cost in large-scale or real-time scenarios.

Most multimodal sentiment analysis models either extract feature vectors from different modalities or perform weighted summation, resulting in the inability to accurately map the data onto a unified multimodal vector space. Moreover, some models rely on image description techniques to convert images into text. These methods extract excessive visual semantic information but fail to precisely capture the sentiment information, leading to information redundancy, increased computational costs, and ultimately affecting the model performance. (22)mentally resolve modal mismatch or the high computational cost of fully aligned attention. These are the problems that need to be urgently solved.

Therefore, we have expored and conducted research on **Dynamic Weighted Gating for Enhanced Cross-Modal Interaction and Fusion** (**DWGCI**) method using. DWGCI is composed of the **Text-Guided Multimodal Fusion Module** (**TGMFM**) and the **Dynamic Gating Module**(**DGM**). Firstly, in TGMFM, it uses text as the query to perform cross-modal attention over visual and other modal streams, thereby aligning them and filtering semantically relevant evidence. After cross-modal interaction, applies learnable, sample-wise gates to dynamically weight each modality, amplifying informative cues while suppressing noise.(22)

Compared with simple concatenation or fully pairwise attention, DWGCI first performs alignment under text and then dynamic weighting, providing robust fusion at a lower computational footprint and mitigating modal mismatch in hateful memes

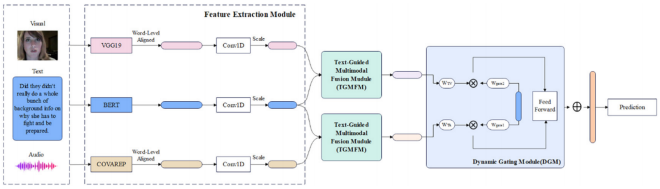



Fig. 8: The overview of DWGCI model

*4) Attention Matrix*

In cross-modal hate meme detection tasks, it is important to understand the semantic relationship between images and texts correctly, particularly in identifying whether the content is hateful (23). In order to build the model of this kind of cross-modal interaction, we have studied and introduced the attention matrix method.

The main goal of the attention matrix is to calculate and show the interaction weights between the images and texts via the cross-attention method. It can provide an intuitive and understandable explanation of the model's decision-making process for content reviewers to understand why the model classifies some image-text pairs as hate speech instead of just giving a binary classification result.

To achieve these goals, the computation of the attention matrix will follow the standard cross-attention mechanism. The purpose of this step is to map the image and text features to the same semantic space so that we can compute their correlation. More specifically, denote image features as and text features as . Then, the queries and keys are obtained by projecting the image and text features, respectively:

$$Q = VW^Q, \quad K = KW^K \tag{14}$$

Then compute the dot product between the query and key and normalize it to get the attention weight matrix:

$$A = \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{E}} + \text{mask}\right) \tag{15}$$

Here, $Q$ and $K$ represent the projections of image and text features, respectively, while the mask ensures proper handling of padding tokens and sequence boundaries. Each element $A_{ij}$ in the matrix $A \in \mathbb{R}^{N \times M}$ quantifies the association strength between the $i$ image region and the $j$.

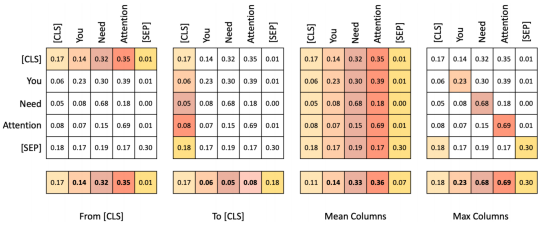



Fig. 9: Interpretation extraction operations from an attention matrix

After getting the base attention matrix, the next goal is to improve the accuracy and trustworthiness of attention explanations. The original attention matrix is spread out among many heads and layers in the model, so using it directly might not give us the best explanations. So we use the Optimus framework proposed by Mylonas et al.. The main idea is to systematically combine the information from all the different heads and layers of attention, and then choose the one that most closely matches the current input.

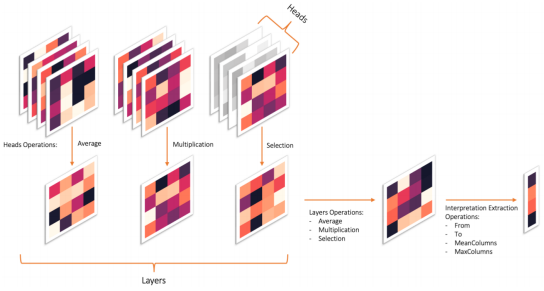



Fig. 10: Head, layer and matrix operations

To assess the quality of various attention-based explanations without supervision and automatically choose the best one, we present the Ranked Faithful Truthfulness (RFT) metric. This metric seeks to gauge the fidelity of explanations by methodically masking input tokens and observing changes in model predictions. The calculation logic is as follows:

$$v(x,z,i) = \begin{cases} f_p(x) - f_p(x^{(-i)}), & \text{if } w_i > 0 \\ f_p(x^{(-i)}) - f_p(x), & \text{if } w_i < 0 \\ -\left|f_p(x) - f_p(x^{(-i)})\right|, & \text{if } w_i = 0 \end{cases} \tag{16}$$

$$\text{RFT}(x,z) = \frac{1}{S}\sum_{i=1}^{S} \frac{v(x,z,i)}{r(t_i)} \tag{17}$$

Although some studies doubt the usefulness of attention-based explanations and recommend using saliency methods instead(24), Mylonas et al.'s experiment(23) shows that attention matrices can give good explanations if they are set up right. On the HateXplain hate speech detection dataset(25),

the BERT-based Optimus method obtained an RFT score of 0.528, which is 193% better than LIME, with 12.5 times faster computation speed.

In summary, by systematically constructing, integrating and evaluating the attention matrix, thishave provided a strong and explainable analysis approach for cross-modal hate speech detection. To clearly separate semantically similar image-text pairs, this gives a solid base for content moderation.

*5) False Claim Module*

Research has shown that hateful content often reinforces prejudice through false claims, which manifest in two primary forms: factual incorrectness, such as referential contradictions between image and text, and deliberate misleading through the distortion of perception using social stereotypes. However, existing methods predominantly rely on the implicit reasoning capabilities of pre-trained models, lacking explicit modeling of falsehood.

To address this limitation, the authors design the FACT module, which incorporates external knowledge via a Social Perception Module to identify misleading content and explicitly captures semantic contradictions through a cross-modal reference graph (26). This approach allows the model to directly target falsehood as a core characteristic of hateful expression. Having discussed the attention matrix in the previous section, we now introduce the Cross-Modal Reference Graph and Graph Neural Network (GNN), which form the core components of the FACT module.

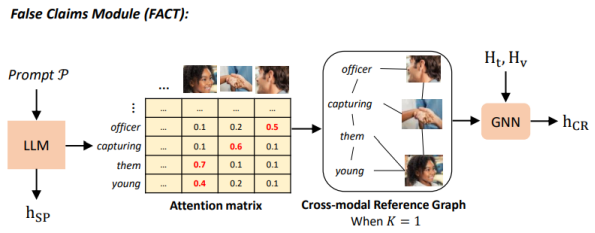



Fig. 11: FACT module

*a) Cross Modual Reference Graph*

The cross-modal reference graph is to capture detailed referential links between visual regions and textual units, allowing the system to pinpoint subtle inconsistencies or conflicts that arise between the two modalities.

The graph is built from two kinds of nodes. They are text tokens and image regions. Their relationships are expressed through three types of edges. These include sequential links within the text, spatial connections across image patches, and cross-modal edges that link specific words to visual areas. The cross-modal links are formed in an adaptive manner. A large language model computes attention weights, and each token is connected to the K image patches that obtain the highest attention from that token.

The edges are generated by relying on the cross-attention signals that arise during multimodal processing. For each token, the model reviews how attention is allocated across all visual regions and keeps only the top-K values. Because the links come directly from the learned attention patterns, no hand-crafted alignment rules are required, and the model can automatically discover associations between fine-grained visual details and the corresponding textual concepts.

Once the graph is assembled, it is processed using a graph neural network for message passing and representation refinement. Each node starts with its respective visual or textual embedding, and several graph convolution layers are applied to propagate information across the structure. A pooling operation is then used to integrate the node features into a single embedding that summarizes the entire graph.

This final representation enables the system to assess whether the textual description is consistent with the visual evidence. For instance, the model can flag mismatches such as a caption mentioning someone "capturing" another person while the image shows a friendly handshake. Such contradictions provide measurable indicators for detecting false or misleading claims.

*b) GNN*

Also, we learned that the graph neural network (GNN) plays a critical role in information aggregation and feature learning within the cross-modal reference graph. Through multi-layer graph convolution operations, the model effectively captures local structural relationships between nodes. The core computational process is expressed as:

$$H^{(l+1)} = \sigma\left(\hat{L}H^{(l)}W^{(l)}\right) \tag{18}$$

Here, $\hat{L}$ represents the normalized graph Laplacian matrix, $H^{(l)}$ denotes the node feature matrix at layer $l$, $W^{(l)}$ corresponds to trainable weight parameters, and $\sigma$ signifies the nonlinear activation function. This formula enables iterative feature propagation along the graph structure, allowing each node to aggregate semantic information from its neighboring nodes.

In implementation, text token nodes and image patch nodes form a heterogeneous graph structure connected through three edge types. After multi-layer graph convolution, each node obtains an enhanced representation incorporating local graph structural information. The graph representation vector is subsequently generated through global average pooling:

$$h_{CR} = \frac{1}{N}\sum_{i=1}^{N} h_i^{(L)} \tag{19}$$

In this function, $N$ indicates the total number of nodes in the graph, and $h_i^{(L)}$ represents the feature vector of the $i$-th node after $L$ graph convolution layers. This pooling operation ensures the resulting graph representation encompasses global cross-modal interaction information.

Theoretical analysis demonstrates that when specific node features undergo significant changes, the resulting graph representation exhibits sufficient variation to satisfy the discriminative condition(26):

$$\|\delta\|_2 > \frac{n|l|}{1^T\hat{L}^{(K)}e_{v_0}\|Pu\|_2} \tag{20}$$

This ensures the model's sensitivity to crucial semantic variations, enabling effective detection of referential inconsistencies between images and text, thereby providing reliable support for false claim identification.

### *D. Potential Applications*

Based on our literature review, we believe memes moderation is a typical multimodal task with significant room for development. We have significantly developed our understanding of the data preprocessing methods and the roles of various modules in the model framework for memes moderation.

From the researches, we knew that current methods face two main challenges: First, the difficulty in effectively detecting implicit hate contents, which often arises from complex interactions between image and text rather than being directly expressed; Second, it is difficult for the model to understand memes in a human-like way, such as interpreting them within a presumed context or connecting them to current hot topics.

Advanced technologies summarized in our review offer good solutions to these issues. For instance, techniques like feature interaction matrices and perceptual encoders can enhance a model's ability to capture fine-grained and contradictory cross-modal semantics, thereby identifying implicit hate speech more accurately. Additionally, technologies like attention matrices and FACT modules we mentioned before can support judgments that better align with human reasoning.

Beyond memes moderation, the technologies discussed above can be applied to other scenarios. For example, in the task of detecting misinformation or fake news on social media, harmful content often leverages the combination of image and text to spread misleading messages. In such cases, an image may be authentic but accompanied by distorted captions, or a real news image may be intentionally placed in an incorrect context. Additionally, certain subtle and emerging forms of insulting language pose challenges for automated models to detect. Addressing these may require integrating knowledge from Large Language Models (LLMs) to enhance semantic understanding and enable more intelligent moderation.

Furthermore, we believe multimodal content understanding holds significant potential in other fields. While our research focuses on semantic alignment and content understanding between image and text, we recognize that these technologies could also be extended to assess the alignment between video and text. This methods might be helpful to contribute to more effective video content moderation.

## III. CONCLUSION

In this survey, we have reviewed recent progress in hateful meme detection from a full-pipeline perspective, moving from data preprocessing to cross-modal reasoning. On the preprocessing side, techniques such as OCR-based text extraction, EAST-driven text-region masking, and Big LaMa inpainting help transform noisy, text-overlaid memes into cleaner visual and textual inputs. Prompt-enhanced networks, LLM-Blender style multi-LLM fusion, and BERT-based keyword extraction further refine the linguistic signal, allowing downstream models to focus on semantically salient and potentially offensive cues. The findings reviewed here underscore that preprocessing forms a vital part of the full detection pipeline, enabling downstream models to operate with greater stability and interpretability.

At the model level, we have investigated a series of encoders and fusion mechanisms. This exploration moves the field from simple modality alignment toward genuine multimodal reasoning. CLIP and specialized perception encoders achieve strong image-text alignment within a shared embedding space. Meanwhile, ROI-based patch extraction builds fine-grained links between textual concepts and local visual evidence. Feature interaction matrices and dynamic weighted gating explicitly model inter-dimensional correlations and modality importance. This allows the system to capture subtle conflicts between images and text. Furthermore, attention matrices and false claim detection modules add a layer of interpretability and external knowledge integration. These components enable the system to make judgments in a more human-like manner. This includes considering real-time news context and learning to make presuppositions.

Our research has laid a solid theoretical foundation for building our system and generated many new ideas. It helped us better recognize existing challenges, such as detecting highly implicit hate speech, keeping up with fast-changing online slang, and reasoning across broader social and temporal contexts. These insights have led us to explore tighter integration of multimodal encoders and large language models, as well as more effective use of external knowledge and real-time information. Overall, this survey has provided clear guidance and strong theoretical support for developing more reliable and explainable hateful content detection systems.

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